

SURAJ KUMAR BISWAS

Software Development Engineer (SDE) | Full-Stack Developer | Cloud & AI
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PROFILE SUMMARY

Software Development Engineer with strong foundations in Data Structures & Algorithms, Object-Oriented Design, and problem solving, with hands-on experience designing and building scalable full-stack applications using Java, Python, React.js, Node.js, Flask, and AWS/Azure. Experienced in designing REST APIs, structuring data models, and building maintainable, scalable software in Agile environments. Built and delivered 10+ production-ready applications, including hackathon-winning and finalist projects at IIT Kharagpur, Hack4Bengal, and IIITM Gwalior. Microsoft Azure Certified (AZ-900), passionate about solving complex engineering challenges and delivering customer-focused software.

WORK EXPERIENCE

Web Development Intern — Edunet Foundation | Remote

Jun 2024 – Aug 2024

- Developed a responsive weather forecasting application using React.js, JavaScript (ES6+), HTML5, CSS3, and REST APIs, delivering a seamless experience across desktop and mobile devices.
- Designed a reusable, component-based frontend architecture and structured API service calls to handle asynchronous data fetching cleanly, improving maintainability and enabling faster feature development.
- Debugged API integration issues including CORS, HTTP errors, and JSON parsing through systematic root-cause analysis, significantly reducing troubleshooting effort.
- Documented API workflows, component architecture, and deployment procedures, improving onboarding efficiency for future contributors.

LEADERSHIP

Mentor — GirlScript Summer of Code (GSSoC) 2024 Extended Edition

Aug 2024 – Nov 2024

Mentored open-source contributors by reviewing pull requests, resolving technical issues, and promoting Git and collaborative software development best practices. Guided participants on project architecture, debugging, and code quality while supporting an active open-source development community.

TECHNICAL SKILLS

- Programming Languages:** Java, C++, Python, TypeScript, JavaScript (ES6+), SQL, C, Django.
- Backend Development:** Spring Boot, Node.js, Express.js, Flask, Microservices, REST API Design, GraphQL, WebSockets, JWT/OAuth, Next.js.
- Frontend Engineering:** Next.js, React.js, Angular, Redux Toolkit, Tailwind CSS, Vite, Responsive Web Design.
- Core CS & Methodology:** Data Structures & Algorithms, System Design, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Agile/Scrum, Git, GitHub, Linux, Complexity Analysis.
- Cloud, Databases & DevOps:** AWS, Microsoft Azure (AZ-900), Docker, PostgreSQL, MongoDB, Supabase, FAISS (Vector DB), CI/CD pipelines.
- Data & Machine Learning:** RAG Systems, LLM Integration, Generative AI, TensorFlow, Keras, OpenCV, Computer Vision, Scikit-learn, Pandas, NumPy, Big Data, LLM API (OpenAI, Groq, or similar)

PROJECTS

AgriNova — AI Smart Agriculture Platform | [GitHub Link](#)

Dec 2025 – Present

React.js, Node.js, Flask, TensorFlow, MongoDB, REST APIs, Gemini API, Google Earth Engine, Express.js.

- Engineered a full-stack AI-powered agriculture platform providing crop disease detection, satellite based crop health detection, weather forecasting and smart farming recommendations, **achieving 92% prediction accuracy while reducing manual diagnosis time by 40%.**
- Designed the MongoDB schema and REST API layer** to support concurrent crop-analysis requests across multiple geographic regions, with Flask microservices separating model inference from application logic to improve fault isolation and horizontal scalability.
- Integrated TensorFlow, OpenCV, and scikit-learn models on the backend with Google Earth Engine datasets to automate and create heatmap and crop health analysis at scale.

VistaLex — Neurodivergent EdTech Platform | [Live Link](#)

Mar 2025 – Jun 2025

React, TypeScript, Vite, Node.js, SQL, TensorFlow, OpenCV, Python

- Developed a full-stack personalized learning platform for students with ADHD, Dyslexia, and Autism using React, TypeScript, Node.js, and Supabase, delivering adaptive educational experiences.
- Integrated TensorFlow, OpenCV, and Python-based computer vision models to enable emotion-aware learning, improving learner engagement by 35% while maintaining a responsive, low-latency UI.

CERTIFICATIONS AND ACHIEVEMENTS

- Microsoft Certified: Azure Fundamentals (AZ-900) | [Credential](#)
- Frontend Developer (React) — HackerRank | [Credential](#)
- Oracle Certified Foundations Associate | [Credential](#)
- Java Programming: Solving Problems with Software — Duke University (Coursera) | [Credential](#)
- Python for Data Science — IBM | [Credential](#)
- Generative AI: Prompt Engineering Basics — Coursera | [Credential](#)
- Remote Sensing and Digital Image Analysis — ISRO | [Credential](#)
- The Joy of Computing Using Python — NPTEL | [Credential](#)
- SQL (Advanced) Certificate — HackerRank | [Credential](#)
- Winner of Prayas 2K25 | April, 2025
- Winner of Hacktopus | July, 2025
- Finalist of East India Blockchain Summit 2.0, IIT Kharagpur | Jan 2026
- Finalist of Hacksagon, IIITM Gwalior | June 2025
- Qualified for State Hub Round, HackFest 2025
- Finalist of Hack4Bengal'25 | June 2025
- Finalist of Hacktropica'25 | June 2025

EDUCATION

B.Tech, Computer Science & Engineering — Techno International Newtown (MAKAUT)

2022 – 2026

CGPA: 7.5/10 | **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, Optimization Mathematics, Distributed Systems, Linux, Algorithm Design, Time Complexity, Space Complexity, Scalability, Fault Tolerance.

Higher Secondary (12th) — Dhakuria High School (WBCHS)

2020 – 2022

Percentage: 77%